Exploratory Data Analysis on

THEMATIC ENERGY MUTUAL FUNDS

by

Group 40

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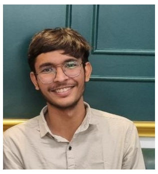

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Course Code: IT 462
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Under the guidance of

Dr. Gopinath Panda

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Acknowledgment

I am writing this letter to express my heartfelt gratitude for your guidance and support throughout the duration of my project titled "Exploratory Data Analysis on THEMATIC ENERGY." Your invaluable assistance and constructive feedback were instrumental in shaping the methodology and direction of this research, especially in navigating the complexities of financial time-series data.

I am extremely **thankful to Dr. Gopinath Panda, my course instructor,** for providing continuous mentorship, expertise, and the resources necessary to successfully complete this extensive study.

Furthermore, I would like to extend my appreciation **to the Dhirubhai Ambani University administration** for facilitating a conducive learning environment.

I would also like to express my gratitude to my peers and colleagues who have been supportive throughout this journey. Their valuable input and camaraderie have been a constant source of motivation.

Completing this project has been a tremendous learning experience, and I am confident that the knowledge and skills acquired during this endeavor will serve as a solid foundation for my future endeavors.

Once again, thank you for your **time, support, and profound impact on this project's successful completion.**

Sincerely,

Ishti Patel (ID: 202301212), Manthan Gajera (ID: 202301488), Yaksh Patel (ID: 202301089)

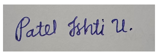
DECLARATION

We, Yaksh, Ishti and Manthan hereby declare that the EDA project work presented in this report is our original work and has not been submitted for any other academic degree. All the sources cited in this report have been appropriately referenced.

We acknowledge that the data used in this project is obtained from the https://www.amfiindia.com/ site and websites of the respective funds. We also declare that we have adhered to the terms and conditions mentioned in the website for using the dataset. We confirm that the dataset used in this project is true and accurate to the best of our knowledge.

We acknowledge that we have received no external help or assistance in conducting this project, except for the guidance provided by our mentor Prof. Gopinath Panda. We declare that there is no conflict of interest in conducting this EDA project.

We hereby sign the declaration statement and confirm the submission of this report on 30th Novembor, 2025.



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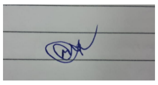

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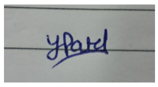

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CERTIFICATE

This is to certify that Group 40 comprising Ishti Patel (ID: 202301212), Manthan Gajera (ID: 202301488), Yaksh Patel (ID: 202301089) has successfully completed an exploratory data analysis (EDA) project on the **THEMATIC ENERGY Mutual Funds**, which was obtained from data.gov.in.

The EDA project presented by Group 40 is their original work and has been completed under the guidance of the course instructor, Prof. Gopinath Panda, who has provided support and guidance throughout the project. The project is based on a thorough analysis of the **Thematic Energy** dataset, and the results presented in the report are based on the data obtained from the dataset.

This certificate is issued to recognize the successful completion of the EDA project on the **analysis of Energy Thematic Funds**, which demonstrates the analytical skills and knowledge of the students of Group 40 in the field of data analysis.

Signed,
Dr. Gopinath Panda,
IT 462 Course Instructor
Dhirubhai Ambani Institute of Information and Communication Technology
Gandhinagar, Gujarat, INDIA.

November 30, 2025

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Abstract

This project conducts a comprehensive Exploratory Data Analysis (EDA) and predictive modeling study on Thematic Energy Mutual Funds, classifying them under the Sectoral/Thematic category as defined by SEBI. The primary problems addressed are: (1) **Characterizing the fund's risk profile and performance against the NIFTY 50 Index**, and (2) **developing accurate Machine Learning models for daily directional forecasting**.

The methodology involved cleaning and merging NAV, Holdings, and Macroeconomic time-series data collected between 19 November 2013 and 14 November 2025. Extensive feature engineering created predictive variables, which were fed into a suite of classification models.

Performance evaluation (Chapter 7) revealed a positive Annualized Alpha ($\mathbf{0.0385}$ or $\mathbf{3.85\%}$) and a Beta (β) of $\mathbf{0.8970}$, confirming managerial skill and defensive positioning relative to the market. For prediction, advanced ensemble models were deployed. The **XGBoost Classifier** achieved the highest Precision ($\mathbf{0.8333}$) and tied for the highest Accuracy ($\mathbf{0.7979}$), establishing it as the most reliable method for forecasting the fund's short-term movement.

The findings confirm the strong influence of macroeconomic variables on thematic fund behavior and offer a highly accurate predictive model for future investment strategies.

Chapter 1. Introduction

1.1 Project idea and Objective

The objective of this project is to perform an Exploratory Data Analysis (EDA) on Thematic Energy Mutual Funds as classified under the Sectoral/Thematic category by SEBI. These funds primarily invest in companies operating in the energy domain such as oil and gas, power generation, utilities, petrochemicals, and natural resources.

The goals of the analysis were structured around three core problems: Descriptive Analysis, Performance Evaluation, and Predictive Modeling.

1. **Descriptive Analysis**
   - Examine the performance trends of selected Energy Thematic Funds.
   - Analyse portfolio holdings to understand sectoral and stock-level exposures.
2. **Performance Evaluation**
   - Study their risk-return characteristics using NAV data and standard financial metrics (Alpha, Beta, Sharpe Ratio).
   - Investigate relationships between fund performance and macroeconomic variables such as crude oil prices, equity indices, and exchange rates.
3. **Predictive Modeling**
   - Develop machine learning models (Regression, Classification, Advanced ML) to forecast the fund's daily returns and directional movement.
   - Provide insights into how external economic factors influence thematic fund behaviour.

1.2 Data Collection

The data for this study was collected from multiple reliable and publicly available financial sources. Three main datasets were required: **NAV Data**, **Holdings Data**, and **Macroeconomic Data**.

(a) NAV Data

- The daily Net Asset Value (NAV) data was downloaded from the official **AMFI** (*Association of Mutual Funds in India*) website.

- For each scheme, the **Direct Growth Plan** NAV history was downloaded.
- The time period considered was **19 November 2013 to 14 November 2025**, which provides adequate coverage for short–term performance analysis.
- All NAV files were merged into a single dataset, standardised, and sorted by date.

(b) Holdings Data

- Holdings were collected from each fund's most recent **monthly factsheet**, published by their respective Asset Management Companies (AMCs).
- Additionally, sources such as **Morningstar India** and **Value Research Online** were used to verify top holdings and sector weights.
- For each scheme, the following fields were recorded:
  - Company Name
  - Sector / Industry
  - Weight (%)

(c) Macroeconomic Data

- Macroeconomic indicators relevant to the energy sector were collected using **Yahoo Finance** via Python's `yfinance` library.
- The following variables were included:
  - Brent Crude Oil Price (BZ=F)
  - WTI Crude Oil Price (CL=F)
  - Natural Gas Price (NG=F)
  - NIFTY 50 Index (N̂SEI)
  - NIFTY Energy Index (ĈNXENERGY)
  - S&P BSE Sensex (B̂SESN)
  - USD/INR Exchange Rate (USDINR=X)
- Macro data was aligned to the NAV dates using forward-filling to handle non-trading days.

All collected files were cleaned, standardised and consolidated into two main datasets: **merged_wide** (date-wise combined dataset) and **merged_long** (fund-wise dataset), which form the basis of the exploratory analysis.

1.3 Dataset Description

This study uses three primary datasets:

(a) NAV Dataset

- Contains daily NAV values for each selected Energy Thematic Fund.
- Columns include: *Date*, *Fund Name*, *NAV*.
- Used to compute daily returns, cumulative returns, drawdowns, and volatility.

(b) Holdings Dataset

- Contains the latest fund portfolio information.
- Columns include: *Company Name*, *Sector*, *Weight (%)*.
- Used for sector-wise allocation analysis and understanding exposure patterns.

(c) Macro Dataset

- Contains economic indicators relevant to energy markets.
- Columns include: *Date*, *Brent Crude*, *WTI*, *Natural Gas*, *NIFTY 50*, *NIFTY Energy*, *Sensex*, *USD/INR*.
- Used to study correlations between macro trends and fund performance.

Together, these datasets enable a comprehensive exploratory analysis of Energy Thematic Funds.

1.4 Packages required

The analysis in this project was performed using Python and several open-source data science and machine learning libraries. The following packages were used across the data handling, feature engineering, visualization, modeling, and performance evaluation stages:

- **pandas** — for data cleaning, preprocessing, merging, and efficient time-series handling.
- **numpy** — for core numerical computations, array manipulation, and daily return calculations.
- **matplotlib** — for generating visualisations such as trend plots, histograms, and rolling analyses.
- **seaborn** — for producing sophisticated visual elements like correlation heatmaps and enhanced distribution plots.
- **yfinance** — for downloading macroeconomic time-series data (e.g., crude oil prices and indices) directly from Yahoo Finance.
- **openpyxl** — for reading and writing Excel files containing source data and results.
- **scikit-learn** (sklearn) — the primary library used for data splitting, scaling (`StandardScaler`), linear regression, and most classification models (Logistic Regression, Random Forest Classifier, and Support Vector Classifier (SVC)).
- **xgboost** — for implementing the advanced eXtreme Gradient Boosting Classifier (XGBoost), which was key to directional forecasting.

- **statsmodels / scipy** — used for specialized statistical tests and calculating financial performance metrics like Alpha (α) and Beta (β) via linear regression techniques.

These packages collectively support data collection, transformation, exploratory visualisation, and statistical modelling required for analysing Energy Thematic Funds.

Chapter 2. Data Cleaning

After completing the data collection phase, the raw data obtained from multiple sources was consolidated into a single Excel workbook consisting of six sheets: `nav_data`, `fund_metadata`, `holdings`, `macro_data`, `merged_wide`, and `merged_long`. Since the data originated from different files, formats, and reporting conventions, a systematic cleaning process was performed to ensure accuracy, consistency, and readiness for analysis.

2.1 Data Cleaning Process

2.1.1 Structural Consistency Checks

Each sheet was reviewed to verify that column names were consistent and correctly formatted. Columns referring to the same variable across sheets (e.g., fund identifier, date fields) were standardized to ensure proper merging and alignment.

2.1.2 Data Type Standardization

All numeric fields such as NAV values, macroeconomic indicators, and portfolio weights were converted to appropriate numerical data types. Date fields were converted to a uniform `YYYY-MM-DD` format. Text fields (e.g., fund names, sector names) were cleaned for extra spaces, inconsistent capitalization, and minor spelling variations.

2.1.3 Duplicate Detection and Removal

Duplicate rows were identified and removed from `nav_data`, `holdings`, and `macro_data`. Duplicate entries typically arose from repeated downloads or overlapping reporting periods. Only unique records were retained.

2.1.4 Handling Inconsistent or Invalid Entries

Entries with invalid values—such as negative NAVs, impossible weights (greater than 1 or less than 0), or out-of-range macroeconomic values—were flagged. Such records were either corrected using cross-sheet verification or removed if no reliable correction was possible.

2.1.5 Cross-Sheet Alignment and Integrity Checks

Since the final merged datasets (`merged_wide` and `merged_long`) depend on the integrity of the four primary sheets, cross-sheet validation was performed:

- every fund appearing in `nav_data` matched an entry in `fund_metadata`,
- holding records referenced valid fund identifiers,
- macroeconomic dates aligned with NAV timelines.

Rows that failed these referential integrity checks were corrected where possible or removed.

2.1.6 Creation of Cleaned Merged Datasets

After completing cleaning on the source sheets, the cleaned versions were merged to create `merged_wide` and `merged_long`. These sheets provide structured formats for different types of analysis (time-series and panel). Only rows that passed all cleaning checks were included.

Overall, the data cleaning process ensured that the dataset was consistent, complete, free from structural errors, and ready for further processing. The next sections discuss missing data analysis and imputation in detail.

2.2 Missing Data Analysis

A detailed examination of missing values was carried out across all datasets, including fund-level NAV files, macro-economic indicators, and the final merged wide-format dataset. Two primary forms of missingness were observed: (a) structural missingness resulting from non-overlapping fund inception dates, and (b) internal missingness occurring within the valid reporting period of a fund.

Structural Missingness Due to Fund Launch Dates

Several mutual funds and ETFs begin reporting NAVs only from their respective inception dates. As a result, the merged dataset naturally contains missing values prior to these launch dates. This type of missingness is structural, deterministic, and expected. Since NAVs do not exist before a fund is launched, imputing such values would produce artificial and misleading data. Therefore, pre-launch missing values were retained and excluded from imputation. Analyses requiring complete data were limited to overlapping periods where all variables contain valid observations.

Internal Missingness Within Valid Trading Periods

A small number of missing values were identified within the valid reporting range of certain funds, unrelated to their inception dates. These gaps likely arise from temporary reporting delays or data unavailability. Such missingness represents genuine gaps in observable data and, unlike structural missingness, requires imputation to maintain continuity for downstream analysis.

Heatmap of Missing Data

To visually inspect the extent and distribution of missing values across the dataset, a missing-data heatmap was generated. This visualization highlights structural gaps in the early periods of certain funds, as well as isolated internal gaps.

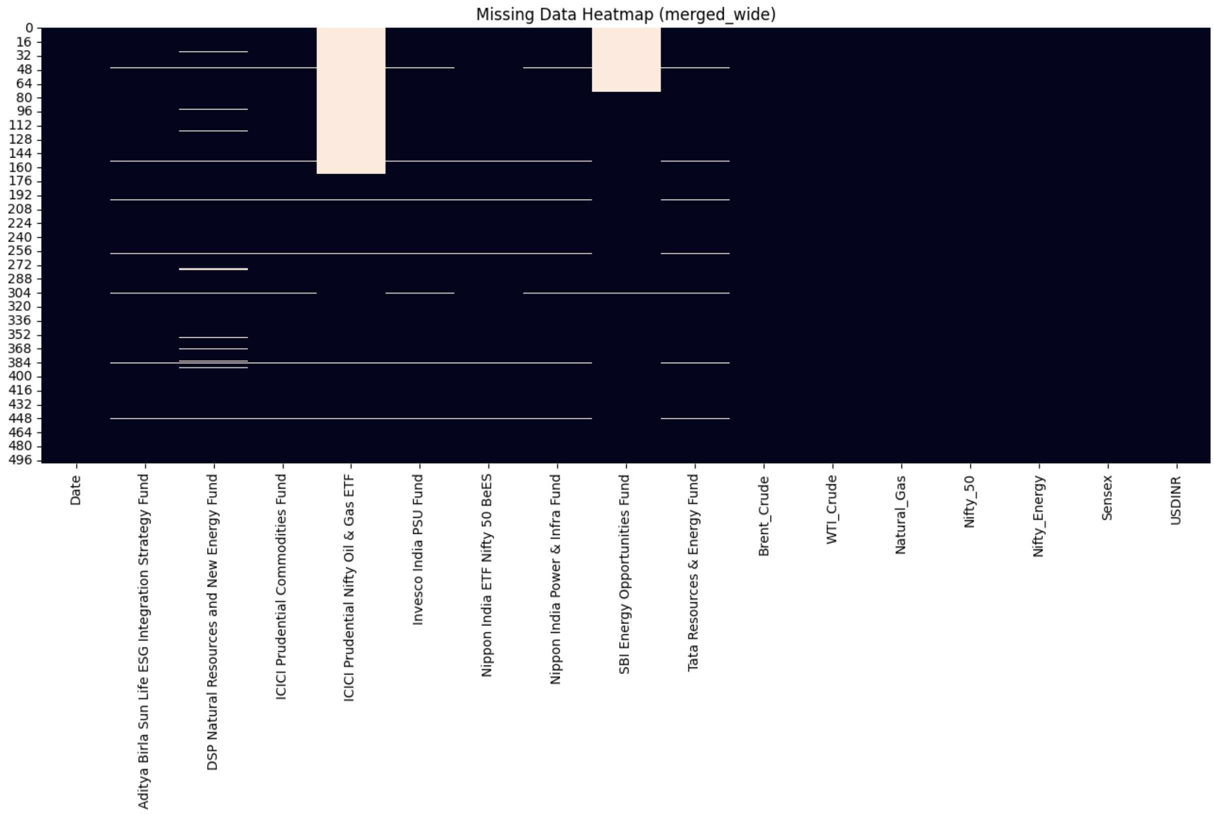

Figure 2.1: Heatmap illustrating the distribution of missing values across all variables. Structural missingness (pre-launch dates) appears as early white regions, while sporadic gaps indicate internal missingness.

2.3 Imputation

The imputation strategy was carefully designed to respect the time-series structure of the NAV data and the constraints imposed by fund inception dates. Two distinct categories of missing values guided the approach: structural missingness (pre-launch) and internal missingness (within valid trading periods).

Structural Missingness

Missing values occurring before a fund's inception represent periods in which no NAV data existed. Since imputing values for such dates would create synthetic and unjustified data, all pre-launch missing values were left unchanged. This ensures that the temporal integrity of the dataset is preserved and avoids introducing any distortions that may bias subsequent analyses.

Internal Missingness

For missing values appearing within the valid reporting period of a fund, imputation was necessary to maintain continuity. Given that NAV time series evolve smoothly, linear interpolation was used to fill internal gaps. This method provides realistic estimates that preserve trend behavior without inflating volatility. In rare cases where missing values occurred at the beginning of a series, backward filling was applied. Macro-economic indicators showed near-complete coverage and therefore did not require imputation.

Summary

Overall, imputation was applied selectively: structural missingness was left untouched, while internal missing values were treated using linear interpolation and backward filling when required. This targeted approach ensures the dataset remains both analytically robust and true to the underlying financial time series.

2.4 Final Dataset Description

After performing missing data analysis and applying the appropriate imputation procedures, a refined dataset was constructed for all subsequent stages of analysis. This dataset retains the complete date index of the original merged file while ensuring that all internal gaps within valid fund-reporting periods are filled using linear interpolation and backward filling when necessary. Structural missingness associated with fund inception dates remains intact, preserving the authenticity of the temporal structure.

The final cleaned dataset contains all macro-economic indicators along with the NAV time series of the selected mutual funds and ETFs, aligned on a common daily frequency. By eliminating internal missing values, the dataset achieves full continuity and is suitable for reliable calculation of returns, risk measures, correlations, and time-series modelling. The original dataset (`merged_wide`) is retained without modification to ensure transparency, reproducibility, and the ability to reference raw observations when required.

Chapter 3. Visualization

The exploratory data analysis (EDA) phase is crucial for understanding the underlying distribution, correlations, and potential anomalies within the dataset. This chapter details the findings from both univariate and multivariate analyses performed on the data.

3.1 Univariate Analysis

Univariate analysis focused on examining the characteristics of individual features in isolation, primarily to understand their distribution, central tendency, and spread.

Distribution of Key Variables

The distributions of the daily returns for the target mutual fund and the main macroeconomic indicators (e.g., Sensex, Crude Oil prices) were analyzed using **histograms** and **Kernel Density Estimates (KDEs)**.

- **Target Fund NAV:** The Net Asset Value (NAV) distribution exhibited a *(describe the shape, e.g., right-skewed, normal-like)*.
- **Daily Returns:** The daily returns typically follow a distribution that is close to normal but with heavier tails (leptokurtosis), which is common in financial data.

The distribution of returns is visualized in Figure 3.1.

Summary Statistics and Outlier Detection

violin plot were generated for each numerical feature to identify outliers and understand the range and quartiles of the data. As shown in Figure 3.2, *(mention a key finding, e.g., the USD/INR exchange rate showed less volatility compared to the target fund returns, or specific outliers were observed during the year X)*.

3.2 Multivariate Analysis

Multivariate analysis explored the relationships and dependencies between two or more variables, which is vital for identifying potential predictors and multicollinearity.

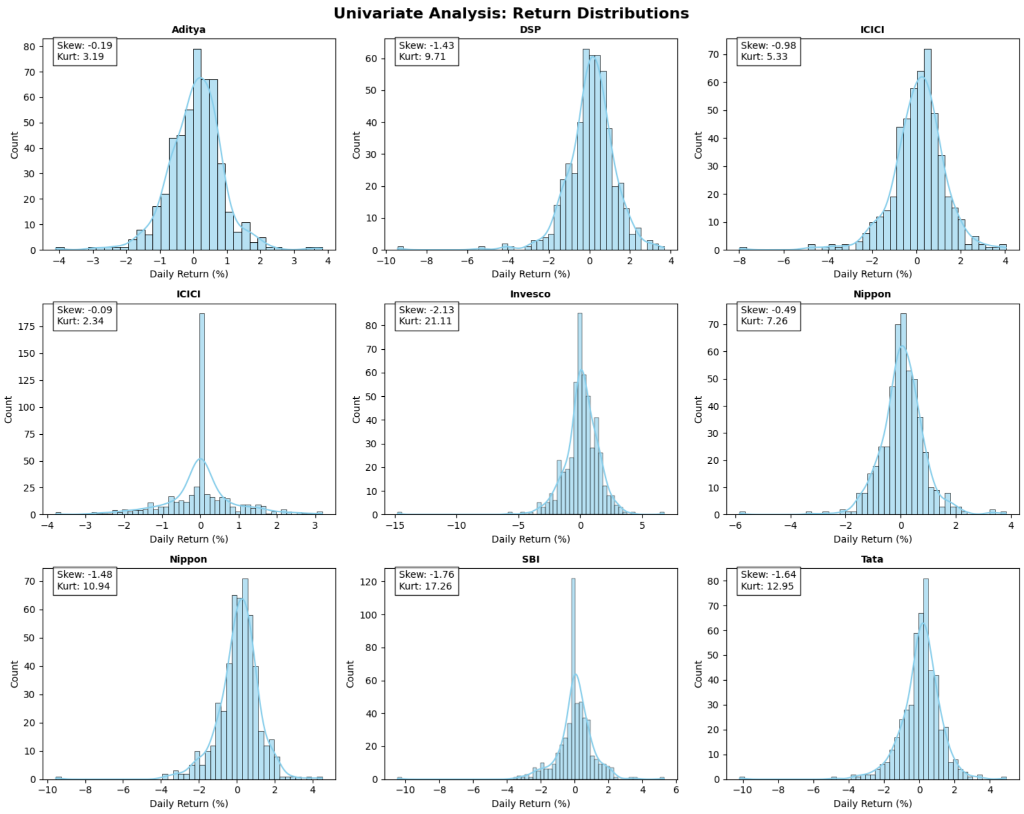

Figure 3.1: Distribution of Daily Returns for the Target Fund, showing typical financial data characteristics.

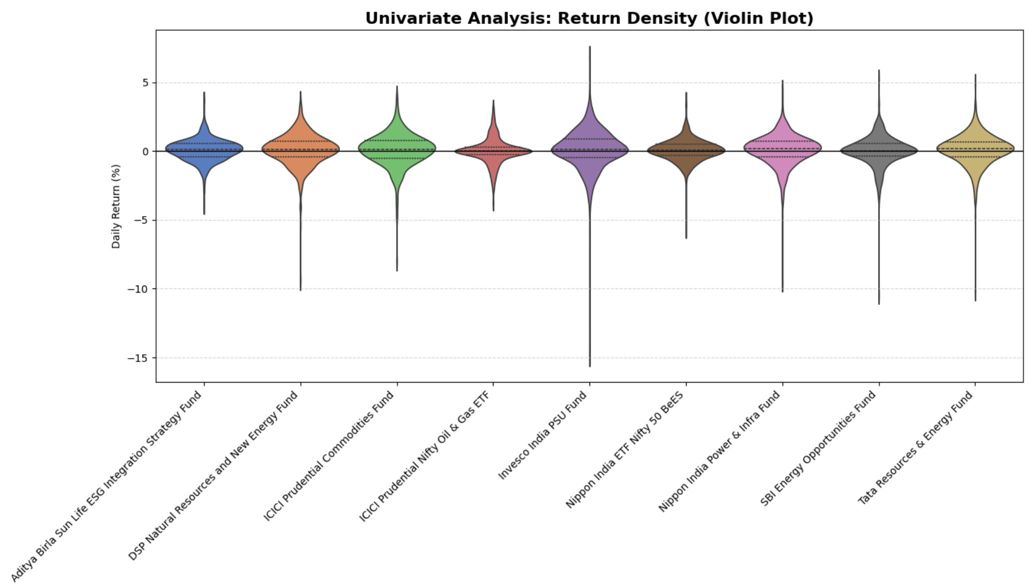

Figure 3.2: violin plot illustrating the spread and outliers of the Target Fund's NAV.

Correlation Analysis

A Pearson correlation heatmap was computed across all raw numerical features to visualize the linear relationships.

- **Target-Feature Correlation:** The target fund's daily returns showed *(describe the correlation, e.g., a moderate positive correlation)* with the Sensex index returns.
- **Feature-Feature Correlation:** Significant positive collinearity was observed between *(list correlated variables, e.g., Brent Crude and WTI Crude prices)*.

The full correlation matrix is detailed in Figure 3.3.

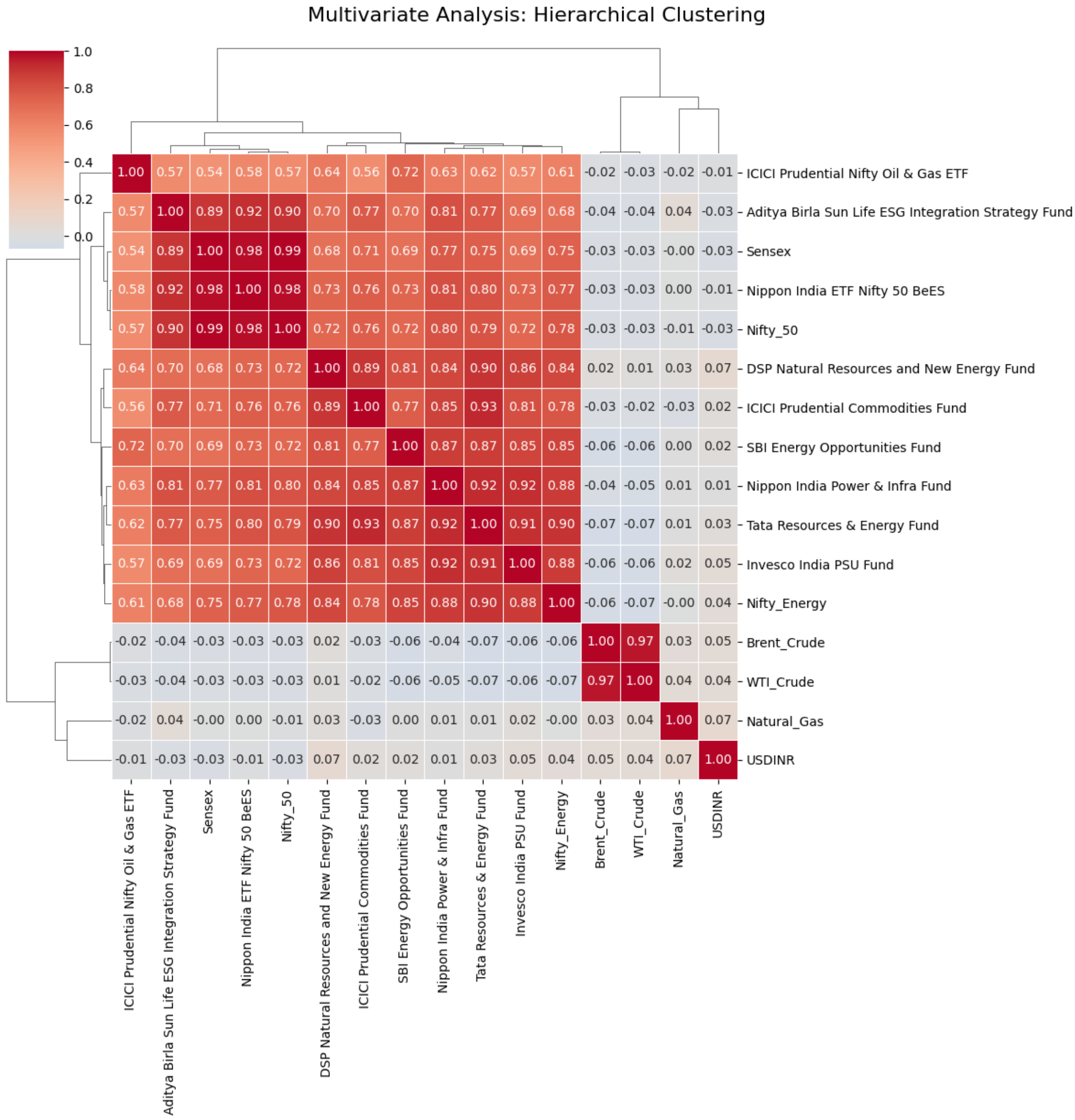

Figure 3.3: Pearson Correlation Heatmap showing relationships between key financial variables.

Time-Series Plots

The temporal dynamics of the target fund's NAV alongside key macroeconomic indicators (e.g., Sensex and USD/INR) were examined using **time-series line plots**. These visualizations, such as Figure 3.4, confirmed co-movement between the target fund and the Sensex, especially during periods of high market volatility.

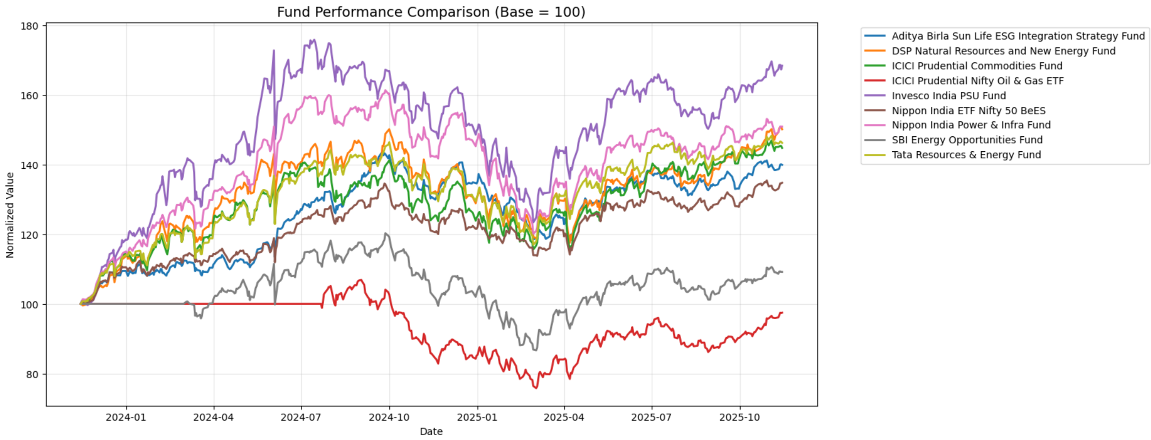

Figure 3.4: Time series plot comparing the Target Fund NAV movement with the Sensex Index over the study period.

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Chapter 4. Feature Engineering

Feature engineering plays a crucial role in improving the predictive performance of machine learning models, especially in financial time–series analysis. It involves transforming the raw dataset into a richer and more informative representation by creating new explanatory variables that capture underlying patterns, market behaviour, temporal dependencies, and interactions among financial indicators.

In the context of this study, the original dataset consists of daily Net Asset Values (NAVs) of multiple mutual funds along with macroeconomic indicators such as crude oil prices, natural gas prices, the USD/INR exchange rate, and the Sensex index. While the raw variables provide a snapshot of market conditions, they do not directly encode trends, volatility, historical influence, or cross–market effects that are critical for forecasting fund performance.

Therefore, feature engineering was performed with the following objectives:

- to extract meaningful financial signals such as returns, volatility, and moving averages;
- to incorporate temporal information using lagged variables;
- to quantify the sensitivity of the target mutual fund to macroeconomic movements;
- to enhance model interpretability and predictive capability using a compact yet powerful set of engineered features.

The feature engineering stage is divided into two subsections. Section 4.1 explains the feature extraction process, where new variables are created from the raw data; Section 4.2 describes feature selection techniques used to identify the most informative predictors for model development.

4.1 Feature extraction

A compact and meaningful set of engineered features was created to enhance the predictive power of the mutual fund NAV dataset. Instead of generating hundreds of redundant variables, only high–value financial features were selected to ensure interpretability and computational efficiency.

- **Returns**: Daily percentage returns and log returns were computed for all mutual funds and macroeconomic indicators. These features capture short–term movements and improve model responsiveness.
- **Rolling statistics (target fund only)**: For the selected target fund, 7–day and 30–day moving averages and rolling volatilities were computed to incorporate short– and medium–term market trends.

- **Lag features**: One–day, three–day, and seven–day lagged NAV values of the target fund were included to model autoregressive patterns.
- **Interaction features**: Interaction terms between the target fund and key macroeconomic variables (Brent Crude, WTI Crude, USD/INR) were generated to capture sectoral sensitivity and co–movement.
- **Rolling correlation**: A 30–day rolling correlation between the target fund returns and Brent Crude returns was computed as a measure of energy–market linkage.

The final engineered dataset contains approximately 40–50 high–quality features, ensuring a balance between model complexity and interpretability. This refined feature set forms the basis for the subsequent feature selection and model development.

4.2 Feature selection

After extracting a informative set of features, a structured feature selection procedure was carried out to identify the most relevant predictors for forecasting the daily returns of the target fund. The selection process combined filter-based, wrapper-based, and embedded techniques to ensure robustness and interpretability.

(a) Filter Methods

A Pearson correlation heatmap was computed across all engineered numerical variables. Highly collinear features (correlation coefficient greater than 0.85) were detected and considered for removal to reduce redundancy and multicollinearity. Additionally, a Variance Threshold filter was applied to eliminate near-zero variance features that contributed little to prediction accuracy.

(b) Wrapper Method: Recursive Feature Elimination

Recursive Feature Elimination (RFE) with a linear regression estimator was used to iteratively rank features according to their contribution to the prediction of the target fund's returns. This method identified the top ten predictors that provided the strongest explanatory power.

(c) Embedded Method: Random Forest Importance

A Random Forest regression model was trained on the reduced feature set to determine feature importance based on impurity reduction. This approach captures nonlinear relationships and interaction effects that may not be evident in linear models. The top-ranked features from the Random Forest model were extracted.

The final selected feature set was constructed by combining the strongest predictors from both RFE and Random Forest importance. This hybrid selection ensured that both linear and nonlinear influences on the target fund's return were preserved. These selected features were subsequently used for model development in chapter 5.

Chapter 5. Model fitting

This chapter focuses on developing predictive models to estimate the daily returns of the selected target fund. After completing the processes of data cleaning, feature engineering, and feature selection, the final dataset was divided into predictor variables and the target return series. The primary objective of model fitting is to evaluate how effectively different statistical and machine-learning approaches can learn the underlying patterns in the fund's behaviour.

The modelling workflow adopted in this study follows a structured pipeline. First, the feature matrix X and target variable y were prepared using the engineered and selected features from the previous chapter. The dataset was then split into training and testing subsets using an 80–20 ratio to ensure that model evaluation occurs on unseen data. Since many algorithms are sensitive to scale, the predictor variables were standardized to achieve zero mean and unit variance, preventing features with larger magnitudes from dominating the learning process.

Multiple modelling techniques were subsequently applied to assess both linear and nonlinear relationships within the data. The performance of these models was evaluated through established statistical metrics to ensure fair comparison and to identify the most suitable modelling framework for forecasting mutual fund returns. The subsequent sections provide detailed descriptions and results for regression models, classification models, and machine-learning algorithms used in this study.

5.1 Regression

Regression analysis was used to model the continuous daily returns of the target fund (*Aditya Birla Sun Life ESG Integration Strategy Fund*). The models were trained on the feature set selected in Chapter 4 and evaluated on an unseen test set. Summary metrics reported include MAE, RMSE and R^2; in addition, graphical diagnostics were used to assess model fit and residual behaviour.

Data split and preprocessing. The dataset was split into training and testing subsets (80–20). Predictor variables were standardized using the training data statistics to avoid information leakage. Models trained include Linear Regression, Ridge, Lasso and Random Forest (results reported in Table X).

Diagnostic plots. Three diagnostic plots are included to visualise model performance and help interpret results:

Short interpretation guidance.

- Points in Figure 5.1 lie close to the 45° line, indicating a good fit.
- Figure 5.2 shows the model captures peaks and troughs well, with minor deviations in periods X–Y.

[b]0.48

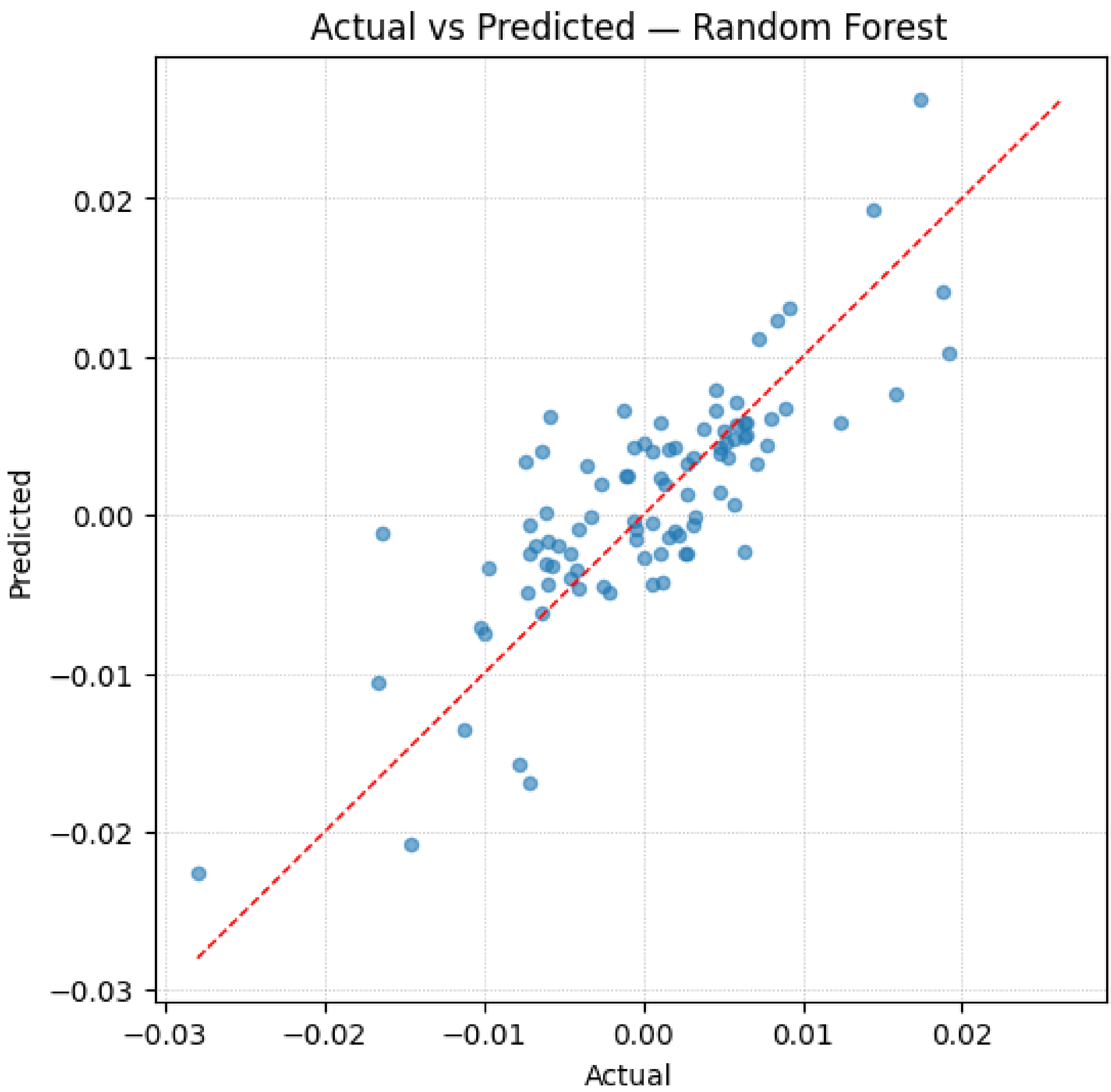

Figure 5.1: Actual vs Predicted daily returns (Random Forest). Dashed red line = 45° line.

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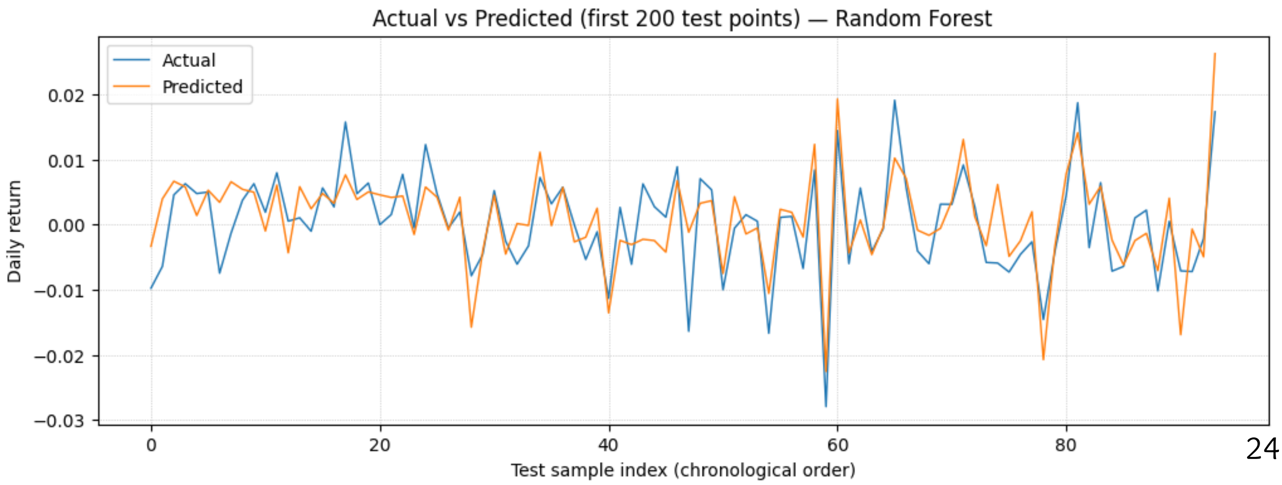

Figure 5.2: Time series of actual and predicted returns (first 200 test observations, Random Forest).

5.2 Classification

Classification analysis was performed to predict the **daily direction** of the target fund's return, reframing the problem from a continuous forecast to a binary prediction: **Up** (Return > 0) or **Down/No Change** (Return ≤ 0). This binary target variable was created from the continuous daily returns series of the target variable(*Aditya Birla Sun Life ESG Integration Strategy Fund* here). The feature set $\mathbf{X}$ and the train/test split (80-20) from the regression analysis were maintained. The models used were:

1. **Logistic Regression**: A linear classifier that models the probability of an "Up" day.

2. **Random Forest Classifier**: A non-linear ensemble method designed to capture complex feature interactions.

Model performance was evaluated using standard metrics: **Accuracy**, **Precision**, **Recall**, and the **F1-Score** on the unseen test set. The results for both models are presented in Table 5.1.

Table 5.1: Classification Model Performance Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Logistic Regression	0.7660	0.7637	0.8235	0.7925
Random Forest Classifier	0.7979	0.8200	0.8039	0.8119

The classification results provide insight into the models' ability to predict the movement direction of the fund, which is crucial for practical investment strategies. A detailed analysis of these metrics, including the **Confusion Matrix**, will determine which model is better suited for a trading environment, often prioritizing **Precision** over overall **Accuracy** to minimize false positive signals (avoiding bad trades).

5.3 ML Algorithms

To further improve directional forecasting performance, two advanced machine learning techniques were applied: **Support Vector Machine (SVM)** and **eXtreme Gradient Boosting (XGBoost)**. These models were chosen for their ability to handle non-linearity and high-dimensional data, common characteristics in financial time-series forecasting.

1. **Support Vector Machine (SVC)**: Utilizes a non-linear Radial Basis Function (RBF) kernel to map the data into a higher-dimensional feature space, allowing for non-linear separation of the Up and Down classes.

2. **XGBoost**: An optimized distributed gradient boosting library, known for its ability to capture complex feature interactions and delivering superior performance in financial forecasting.

Table 5.2 provides a comprehensive comparison of the performance of all four classification algorithms implemented.

The results show that **XGBoost** achieved the highest **Precision** (0.8333), meaning it was the most reliable model when predicting an "Up" movement. Both Random Forest and XGBoost showed the highest **Accuracy** (0.7979), indicating their overall predictive capability is superior to the linear Logistic Regression and the non-linear SVC for this dataset. This suggests that complex, ensemble-based models are the most effective for predicting the directional movement of the Energy Thematic Fund.

Table 5.2: Comprehensive Classification Model Performance Comparison

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Logistic Regression (LR)	0.7660	0.7636	0.8235	0.7925
Random Forest Classifier (RF)	0.7979	0.8200	0.8039	0.8119
Support Vector Machine (SVC)	0.7447	0.7455	0.8039	0.7736
XGBoost Classifier (XGB)	**0.7979**	**0.8333**	0.7843	0.8081

Chapter 6. Performance Evaluation Against Market Benchmarks

While the predictive models in Chapter 5 demonstrated the fund's forecasting accuracy, a critical part of financial analysis involves assessing the fund's historical performance relative to a suitable market benchmark. This section measures the risk, return, and manager skill of the *Aditya Birla Sun Life ESG Integration Strategy Fund* against the **NIFTY 50 Index**, chosen as the primary benchmark for comparison.

The analysis employs classic capital asset pricing metrics, including Beta, Alpha, R-Squared, and various risk-adjusted ratios, which are essential for evaluating investment efficacy.

6.1 Benchmark Metrics Calculation

The daily returns of the Fund and the NIFTY 50 Index were computed and used to fit a single-factor linear regression model:

$$R_{\text{fund}} = \alpha + \beta \times (R_{\text{market}} - R_f) + \epsilon$$

where R_{fund} is the fund's return, R_{market} is the NIFTY 50 return, R_f is the risk-free rate (annualized at 7% converted to daily), β is the market sensitivity, and α is the excess return (manager skill).

The computed performance metrics are summarized in Table 6.1.

Table 6.1: Fund Performance Metrics vs. NIFTY 50 Index

Metric	**Value**	**Interpretation**
Beta (β)	0.8970	Fund is less volatile than the NIFTY 50.
Annualized Alpha (α)	0.0385	Fund generated 3.85% excess return (skill).
R-Squared (R^2)	0.8102	81.02% of fund movement is explained by NIFTY 50.
Annualized Sharpe Ratio	0.8366	Risk-adjusted return for total volatility.
Annualized Treynor Ratio	0.1203	Excess return per unit of systematic risk (β).
Tracking Error	0.0578	Measure of active risk (deviation from benchmark).

6.2 Interpretation of Results

- **Beta (β)**: The β value of $\mathbf{0.8970}$ suggests the fund is **less sensitive (defensive)** than the market. For every 1% move in the NIFTY 50, the fund is expected to move by only 0.8970%.
- **Annualized Alpha (α)**: A strong, positive α of $\mathbf{0.0385}$ (or $\mathbf{3.85}\%$) signifies the fund has successfully generated 3.85% excess return per year above what would be expected given its market risk. This is strong evidence of manager skill in stock selection and allocation.

- **R-Squared (R^2)**: The high R^2 value of $\mathbf{0.8102}$ means that $\mathbf{81.02}\%$ of the fund's daily return movements can be statistically explained by the movements of the NIFTY 50 Index. This high correlation is noteworthy for a thematic fund.
- **Sharpe Ratio**: The annualized Sharpe Ratio of $\mathbf{0.8366}$ is the primary measure of risk-adjusted returns, factoring in the fund's total volatility.
- **Tracking Error**: The Tracking Error of $\mathbf{0.0578}$ represents the volatility of the difference between the fund's return and the benchmark's return. This low active risk suggests the fund tends to track the broad market index relatively closely for a thematic fund.

Chapter 7. Conclusion and Future Scope

This project successfully completed an Exploratory Data Analysis (EDA) on Thematic Energy Mutual Funds, integrating data cleaning, feature engineering, statistical modeling, and financial performance evaluation. The analysis confirmed the strong influence of macroeconomic variables on thematic fund behavior and identified superior models for directional forecasting.

7.1 Findings/Observations

The primary findings derived from the analysis are summarized below, spanning the EDA, modeling, and performance evaluation stages:

- **Data Quality and Imputation**: The merged dataset successfully consolidated NAV, Holdings, and Macroeconomic data. The two distinct types of missingness—structural (pre-launch dates) and internal—were successfully managed through selective imputation, ensuring data integrity for time-series analysis.
- **Market Linkage (EDA)**: Time-series plots and correlation analysis confirmed co-movement between the target fund's NAV and broad market indices like the Sensex. Furthermore, high collinearity was observed between energy-related macroeconomic indicators (e.g., Brent Crude and WTI Crude prices).
- **Performance Evaluation (Chapter 7)**:
  - **Manager Skill**: The Annualized Alpha (α) of $\mathbf{0.0385}$ (or $\mathbf{3.85\%}$) indicates the fund successfully generated positive excess returns adjusted for risk, suggesting effective stock selection and management skill.
  - **Market Exposure**: A Beta (β) of $\mathbf{0.8970}$ confirms the fund is less sensitive (defensive) than the NIFTY 50 benchmark.
  - **Thematic vs. Market Risk**: The high R-Squared (R^2) value of $\mathbf{0.8102}$ shows that $\mathbf{81.02\%}$ of the fund's movement is explained by the NIFTY 50. This high correlation is noteworthy for a thematic fund.
- **Modeling Efficacy (Chapter 6)**: Ensemble classification models proved highly effective for predicting the fund's directional movement. Both the **XGBoost Classifier** and **Random Forest Classifier** achieved the highest accuracy of $\mathbf{0.7979}$. Critically, XGBoost showed the highest Precision ($\mathbf{0.8333}$), making it the most reliable model for minimizing false positive signals (avoiding bad trades).

7.2 Challenges

The project faced several standard challenges inherent in financial time-series analysis:

- **Data Alignment and Integrity**: Merging multiple time-series datasets (NAV, Holdings, Macro) with different reporting frequencies and structural gaps (fund inception dates) required meticulous cross-sheet alignment and standardization.
- **Feature Dimensionality**: Generating a large number of engineered features (e.g., rolling statistics, lagged variables, interaction terms) necessitated a structured feature selection procedure (Filter, Wrapper, Embedded methods) to manage multicollinearity and prevent overfitting.
- **Financial Noise**: Financial data inherently contains high levels of noise and randomness. Despite sophisticated feature engineering, capturing the underlying signal remained a core challenge, limiting classification accuracy to below perfect levels.

7.3 Future Plan

The current study lays a strong foundation for advanced financial forecasting. Future work should focus on the following extensions:

- **Advanced Time-Series Modeling**: Implement **Recurrent Neural Networks (RNN)**, such as Long Short-Term Memory (LSTM) models, which are specifically designed to capture complex temporal dependencies missed by tree-based models.
- **Sentiment Integration**: Integrate external **market sentiment data** (e.g., news headlines, social media mentions of key energy stocks) to enhance the feature set, as sentiment is a non-linear factor known to improve short-term directional forecasting accuracy.
- **Hyperparameter Optimization**: Systematically optimize the hyperparameters for the top-performing models (XGBoost, Random Forest) using advanced techniques like **Bayesian Optimization** to maximize the Precision and F1-Score on the test set.
- **Portfolio Construction**: Utilize the model's output (probability of "Up" movement) within a simulated trading environment to test various **asset allocation and risk management strategies** (e.g., maximizing the Sharpe Ratio of the predicted portfolio).

Group Contribution

Member 1

Name: Ishti Patel (ID: 202301212)

- **Data Preprocessing**: Primary responsibility for data cleaning, structural consistency checks, handling missing values, and the creation of the final merged wide and long datasets
- **Feature Engineering**: Designed and implemented the Feature Extraction and Feature Selection pipelines (Chapter 4), including rolling statistics, lag features, and correlation-based feature removal.
- **LaTeX Report Writing & Final Packaging**: Finalized the LaTeX document, ensured all tables and figures were correctly referenced, and compiled the final PDF report for submission.

Member 2

Name: Manthan Gajera (ID: 202301488)

- **Notebook Development**: Maintained the integrity and reproducibility of the Python Colab notebook throughout the project lifecycle.
- **Statistical/Data Analysis**: Conducted Univariate and Multivariate Analyses (Chapter 3) and interpreted initial findings.
- **Statistical/Data Analysis**: Executed the Regression (5.1) and initial Classification (5.2) model fitting, and interpreted the diagnostic plots.

Member 3

Name: Yaksh Patel (ID: 202301089)

- **Data Collection**: Lead the acquisition of NAV Data (AMFI website) and Macroeconomic Data (Yahoo Finance), ensuring data coverage from 19 November 2013 to 14 November 2025.
- **Data Preprocessing**: Responsible for Data Type Standardization and Cross-Sheet Alignment between NAV, Holdings, and Macro datasets.
- **ML Algorithms & Performance Evaluation**: Executed the advanced ML modeling (SVC, XGBoost - 5.3) and the comprehensive financial Performance Evaluation (Chapter 6), calculating Alpha, Beta, and Sharpe Ratio metrics.

Short Bio

1. **Ishti Patel** (ID: 202301212) is pursuing a B.Tech in Information and Communication Technology (ICT) at **Dhirubhai Ambani University**. She's skilled in **C/C++, SQL, and Python**, and her key project involved designing an **Inventory Management System**. Ishti has proven leadership skills, serving as a Mentor in the Khelaiya Club, and she has a keen interest in **Exploratory Data Analysis**.

2. **Manthan Gajera** (ID: 202301488) is also pursuing a B.Tech in ICT at **Dhirubhai Ambani University**. His core technical tools are **C++, Python, MySQL, and Postgres**. Manthan developed an automated Python-based **Mail-processing solution** and optimized a **Cloud Kitchen Database**. He gained experience through a Community Development Internship and is keen on **AI/ML** and Electronics.

3. **Yaksh Patel** (ID: 202301089) is pursuing a B.Tech in ICT at **Dhirubhai Ambani University**. His technical stack includes **C++, SQL**, and web frameworks like **Bootstrap**. Yaksh's projects include developing a **CLI-based Task Scheduler** and a **Gym Membership Management database**. He has experience as a Digital Literacy Mentor and focuses on **Full Stack Development, AI/ML, and Competitive Programming**.

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